



## Sets

### Python Skills Module 2: Topic 9

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## Practical Skills Module 2: Topic 9

|            |                                                              |
|------------|--------------------------------------------------------------|
| Topic Code | PM-02-PS09                                                   |
| Topic      | Use Python sets to store multiple items in a single variable |

### ***Scope of Practical Skill***

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0901 Create Python sets
- PA0902 Add or remove elements
- PA0903 Sort lists
- PA0904 Determine how many items a set has
- PA0905 Join sets

### ***Applied Knowledge***

- AK0901 Concept, definition and functions of Python data types

### ***Internal Assessment Criteria***

- IAC0901 Expected results with Python sets are achieved

### **Resources:**

| Knowledge                     | Information from Python Programmer Curriculum |
|-------------------------------|-----------------------------------------------|
| National Curriculum Framework | 900221-000-00-PM-02                           |



### **Practical task**

Follow the facilitator's instructions to complete the following activities:

### Scenario: Managing a Library Book Collection

In this scenario, a library manager uses Python to manage a collection of book genres available in the library. The program will use sets to track genres, perform operations like adding or removing genres, sorting lists of books within a genre, checking the number of available genres, and merging multiple genre sets when new collections are acquired.

Create a Python file called **genre\_manager.py** and complete the following steps:

**Step 1:** Create sets for available book genres.

- Create two sets:
  - One for **fiction genres** (e.g., 'Fantasy', 'Romance', 'Historical')
  - One for **non-fiction genres** (e.g., 'Biography', 'Science', 'Self-help')
- Store them in clearly named variables.

*Tip:* Sets are defined using curly braces {} and contain unique, unordered items.

**Step 2:** Add and remove elements from sets.

- Add the genre "Thriller" to the **fiction set** using an appropriate set method.
- Remove the genre "Science" from the **non-fiction set**.
- Print both sets to show the updated collections.
  - *Expected output:*  
Updated fiction genres: {'Fantasy', 'Romance', 'Historical', 'Thriller'}  
Updated non-fiction genres: {'Biography', 'Self-help'}

**Step 3:** Sort a list of books in a genre.

- Create a list of books in the **Fantasy** genre:
  - "The Hobbit", "Harry Potter", "Game of Thrones"
- Sort the list alphabetically and print the sorted result.
  - *Expected output:* Sorted Fantasy Books: ['Game of Thrones', 'Harry Potter', 'The Hobbit']

*Note:* This step uses a list, not a set, because sets cannot be sorted.

**Step 4:** Count the number of genres in each set.

- Use the `len()` function to determine how many genres are in:
  - The fiction set
  - The non-fiction set
- Print the results with appropriate labels.
  - *Example output:*  
Number of fiction genres: 4  
Number of non-fiction genres: 2

**Step 5:** Merge fiction and non-fiction genres.

- Combine the two sets to form a complete set of **all genres available** in the library.
- Use a set operation (e.g., `union()`) to merge them into a new set.
- Print the final combined set.
  - *Expected output:*  
All available genres: {'Fantasy', 'Thriller', 'Biography', 'Romance', 'Historical', 'Self-help'}



### Take note

**Important:** Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.

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